

MAT236 - Homework 4 (SENT ON 29.04.2026)

Q1. Let $z = x + iy$, where $x, y \in \mathbb{R}$. Using the exponential definition:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Substitute $z = x + iy$:

$$\cos(x + iy) = \frac{e^{i(x+iy)} + e^{-i(x+iy)}}{2} = \frac{e^{ix-y} + e^{-ix+y}}{2} = \frac{e^{-y}e^{ix} + e^ye^{-ix}}{2}$$

Now use Euler's formula:

$$e^{ix} = \cos x + i \sin x, \quad e^{-ix} = \cos x - i \sin x$$

Substitute:

$$f(z) = \frac{e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)}{2}$$

Expand:

$$f(z) = \frac{(e^{-y} + e^y) \cos x + i(e^{-y} - e^y) \sin x}{2}$$

Use hyperbolic identities:

$$\cosh y = \frac{e^y + e^{-y}}{2}, \quad \sinh y = \frac{e^y - e^{-y}}{2}$$

So:

$$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$$

$\operatorname{Re}\{\cos z\} = \cos x \cosh y, \quad \operatorname{Im}\{\cos z\} = -\sin x \sinh y$
